AMERICAN ASSOCIATION *of* PHYSICISTS IN MEDICINE

# 2026 SUMMER SCHOOL

Transforming Technology, Modern Practice, and Clinical Impact

**JUNE 16–20** | UNIVERSITY OF MICHIGAN

# Image Quality Considerations

June 18th | 10:30-11:00am


Jess Scholey PhD DABR
University of California San Francisco



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# Disclosures

Research funding from

Siemens Healthineers and Varian Medical Systems

# Disclosures

My expertise: MRI for radiation therapy

**BIG THANKS** to

Dennis Stanley, PhD

CBCT-ART Expert

THE UNIVERSITY OF ALABAMA AT BIRMINGHAM

Huixiao Chen, PhD

BgRT-ART Expert

Yale University School of Medicine

# Goals

1. Understand primary image quality considerations for **contouring**, **dose calculation**, and **image registration**.
2. Review image quality considerations for:
   - MR-guided ART
   - CBCT-guided ART
   - PET-guided ART

# Image Quality Considerations

Image quality requirements depend on the application.

# General workflow of ART

cancers

MDPI

*Review*

**Adaptive Radiotherapy: Next-Generation Radiotherapy**

Olga Maria Dona Lemus [1], Minsong Cao [2], Bin Cai [3], Michael Cummings [1] and Dandan Zheng [1,*]

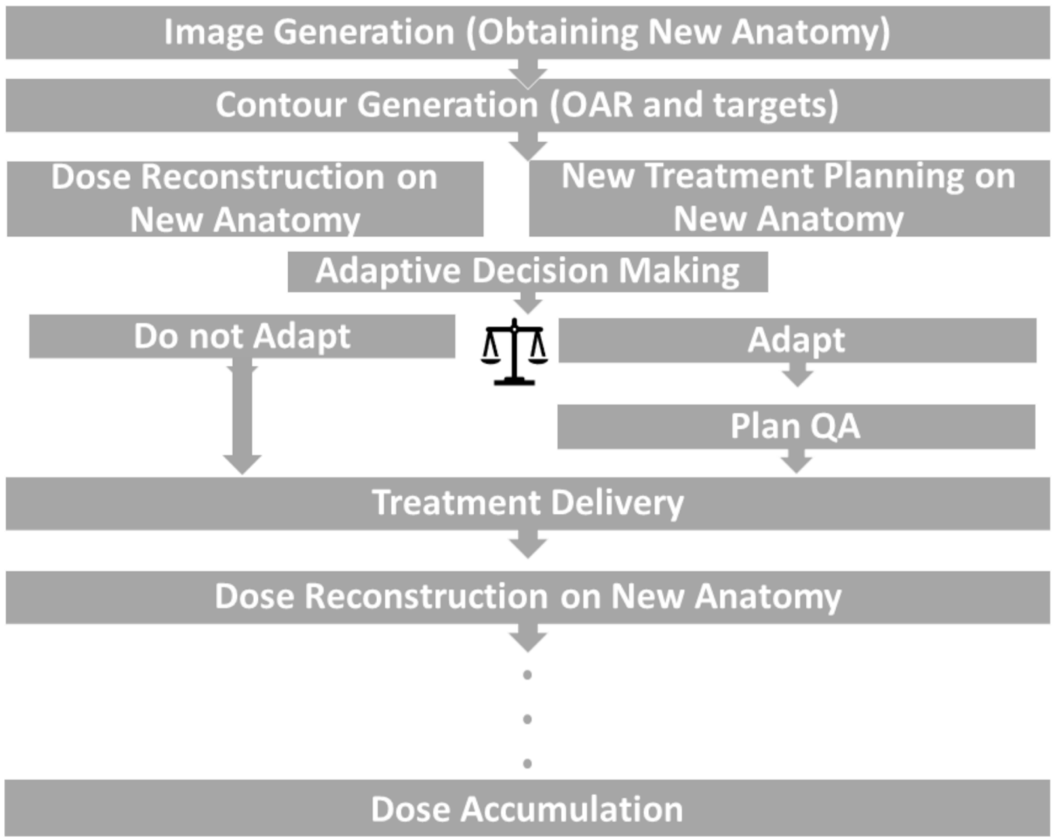


**Figure 2.** General workflow of ART.

# General workflow of ART

cancers MDPI

*Review*

**Adaptive Radiotherapy: Next-Generation Radiotherapy**

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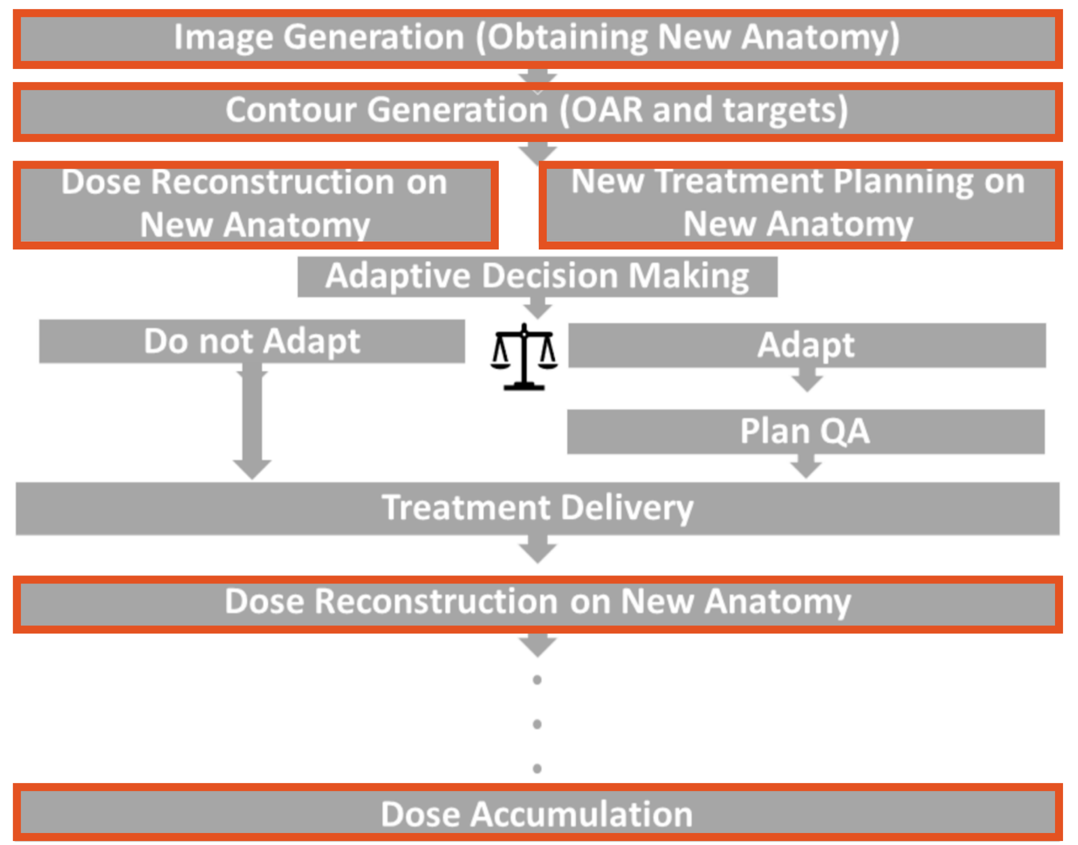



**Figure 2.** General workflow of ART.

# General workflow of ART

## Image Generation

- Total Session Time ~30-45 mins.
- Image acquisition and reconstruction should be minimized
  - i.e. seconds or minutes.

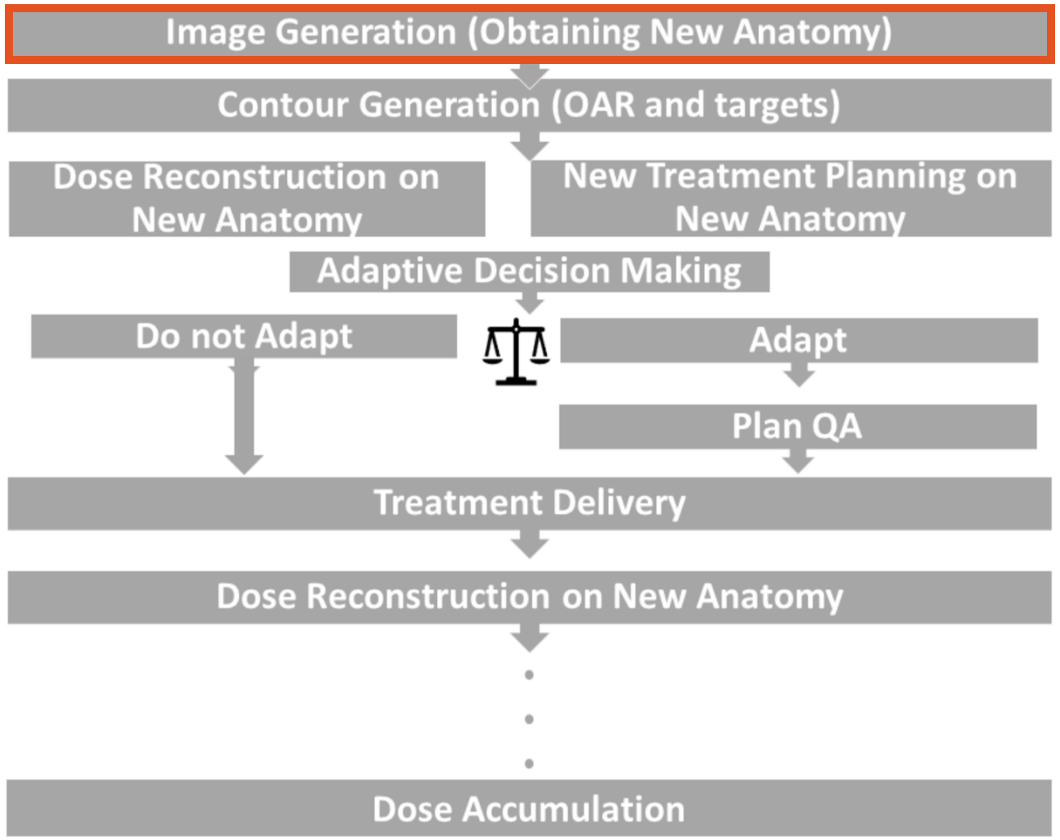



**Figure 2.** General workflow of ART.

# General workflow of ART

## Contour Generation

- Image quality must be suitable for delineation of targets and organs-at-risk (OARs).
  - Manual and auto-contouring.
  - Image registration to propagate contours from planning data.

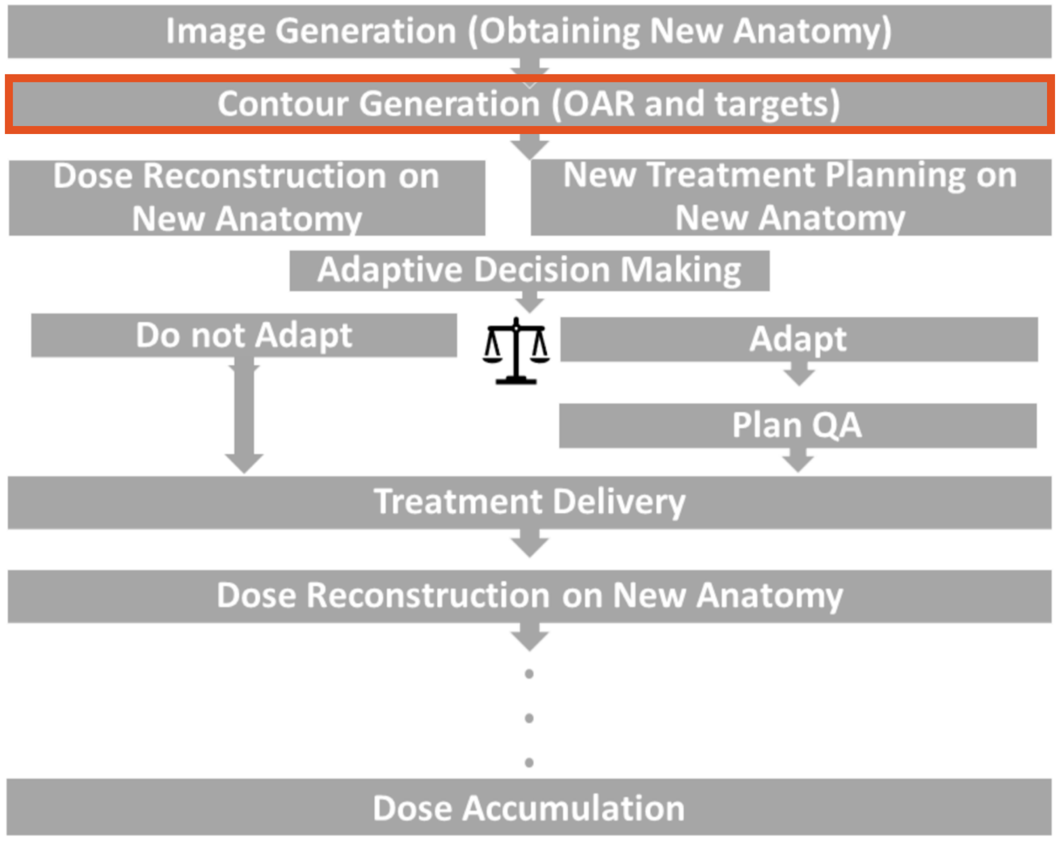



**Figure 2.** General workflow of ART.

# General workflow of ART

## Dose reconstruction on New Anatomy

- Images must provide accurate electron density for dose calculation.
- May prompt decision on offline ART (e.g. proton therapy).

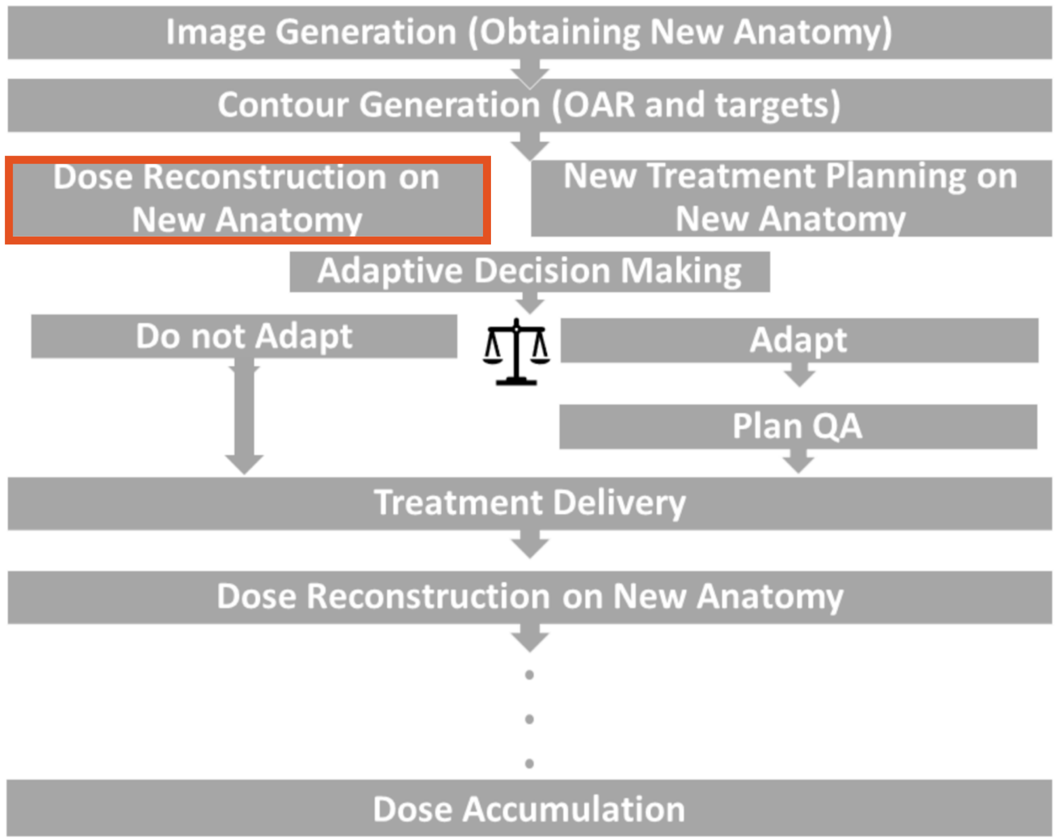



**Figure 2.** General workflow of ART.

# General workflow of ART

## Dose optimization

- Images must provide accurate electron density for dose optimization and replanning.
- Goal to ensure:
  - Overall dosimetric uncertainty ±5%.

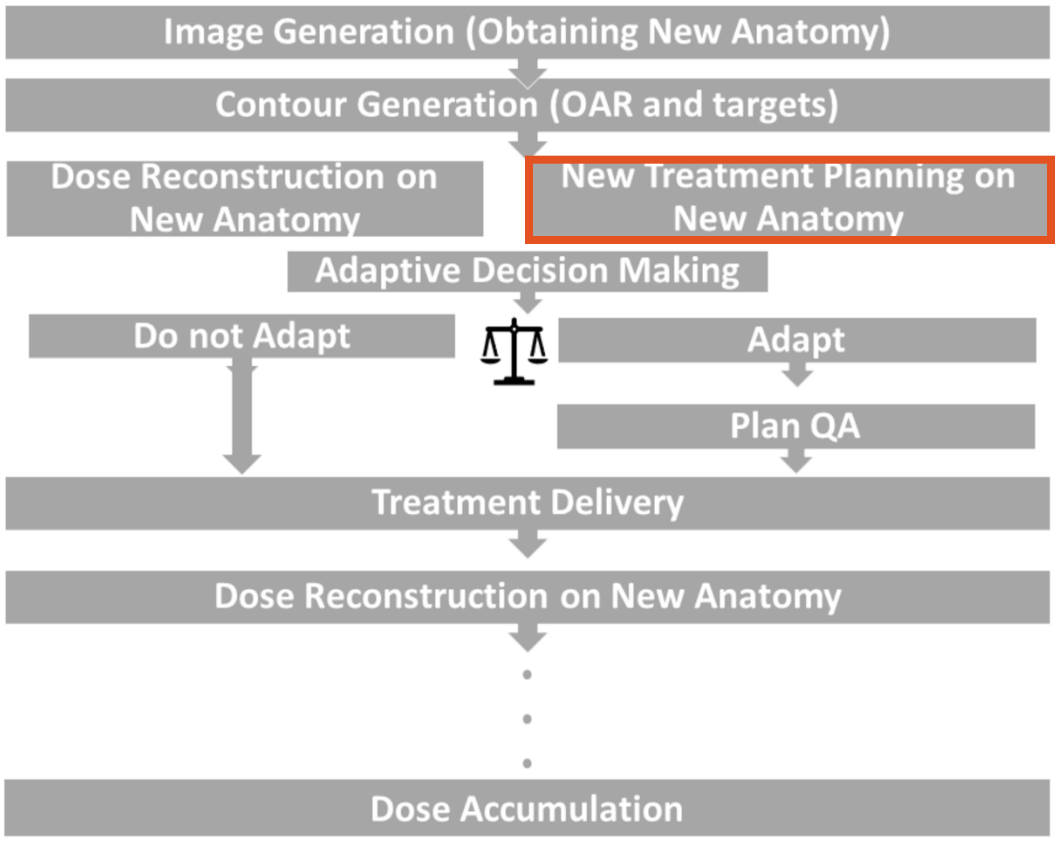



**Figure 2.** General workflow of ART.

# General workflow of ART

## Dose Accumulation

- Image quality must be adequate for co-registration to facilitate dose accumulation across ART course.
- Accuracy can be compromised by:
  - Poor quality of input images (planning and daily in-room images)
  - Presence of artifacts and distortion.

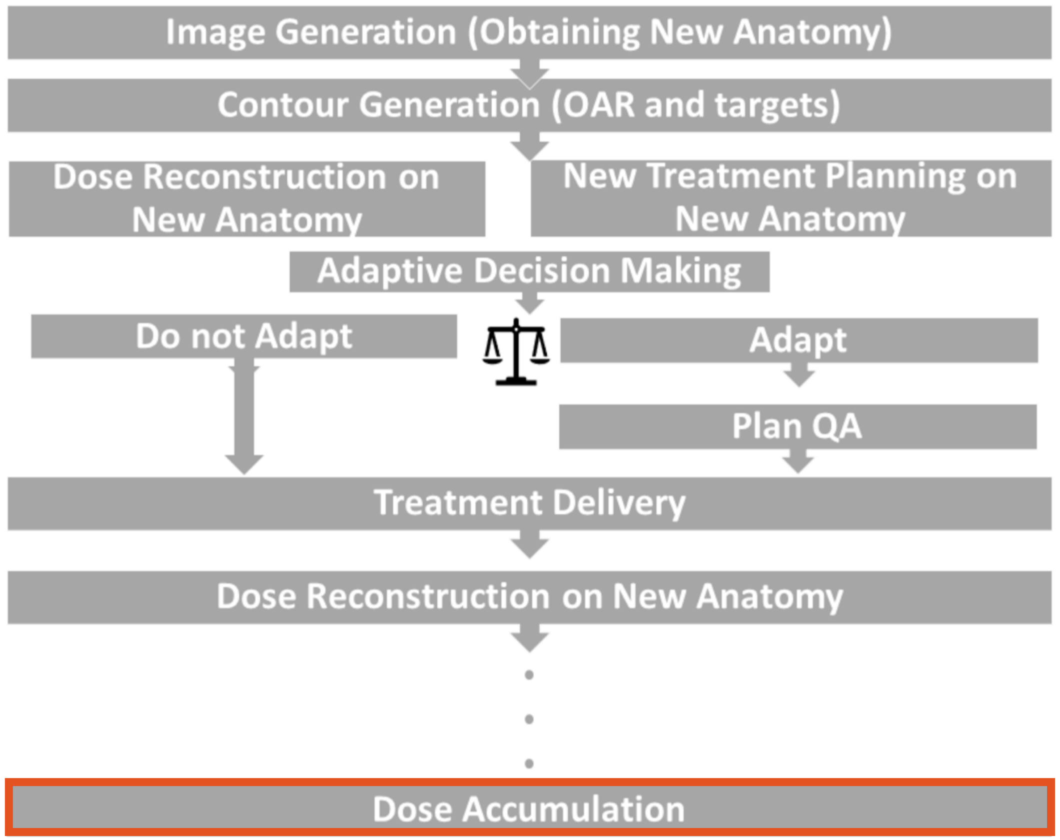



**Figure 2.** General workflow of ART.

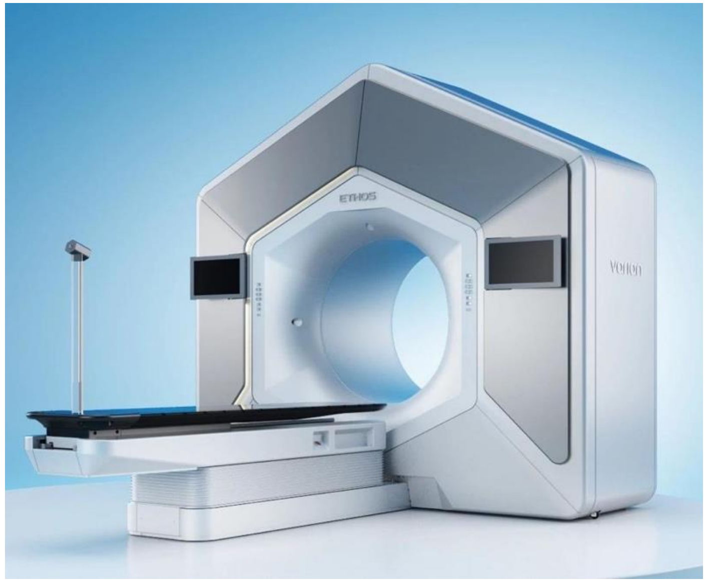
# Other considerations

- Geometric accuracy
  - Recommended tolerance:
    - 1mm for stereotactic RT
    - 2mm for non-stereotactic RT
- Motion artifacts
- Metal, air artifacts

# Imaging Used in (online) ART

MRI-based


CBCT-based




BgRT-based

# Imaging Used in (online) ART

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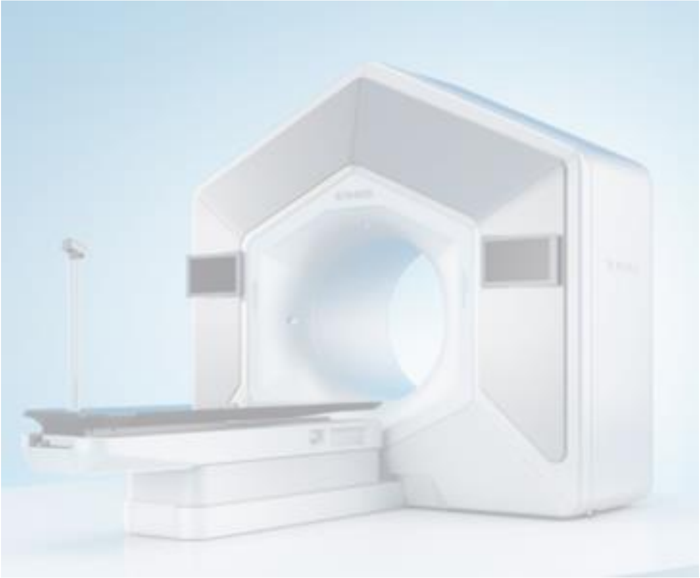


BgRT-based

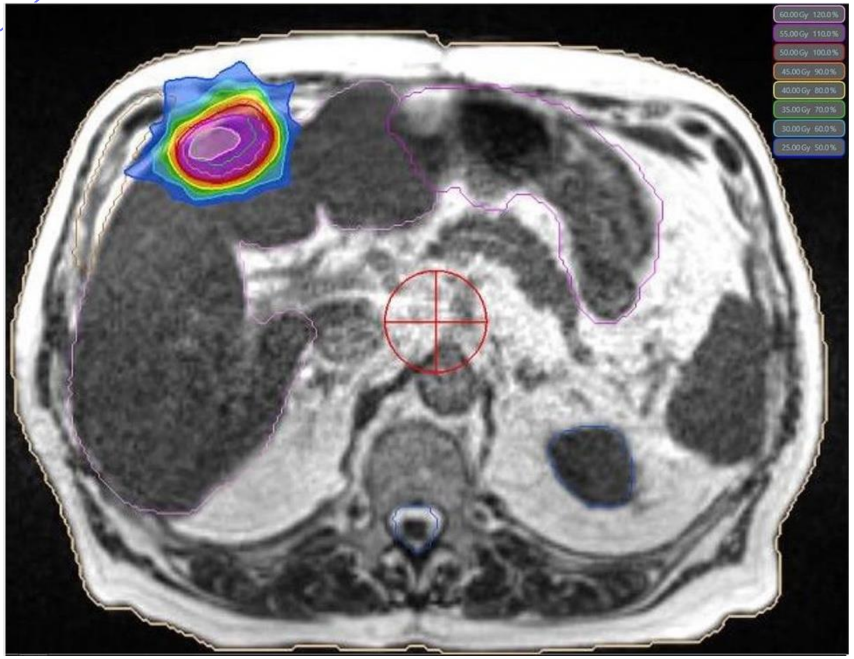
# Contouring

MRI-based

- ✓ **Contouring**
- ✓ Dose calculation
- ✓ Image registration
- ✓ Artifacts

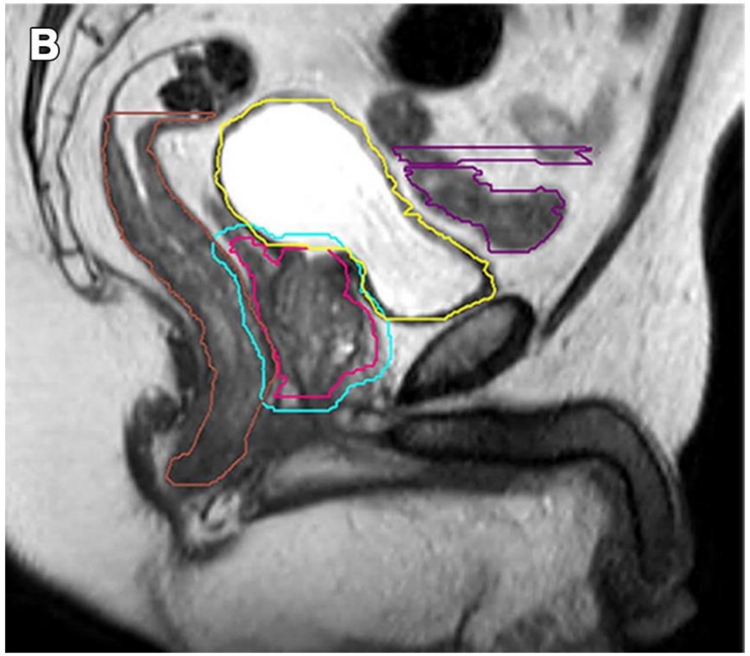
- MRI demonstrates excellent soft tissue contrast.
  - T1w and T2w images used for contouring target and OARs.

**Liver Tumors**



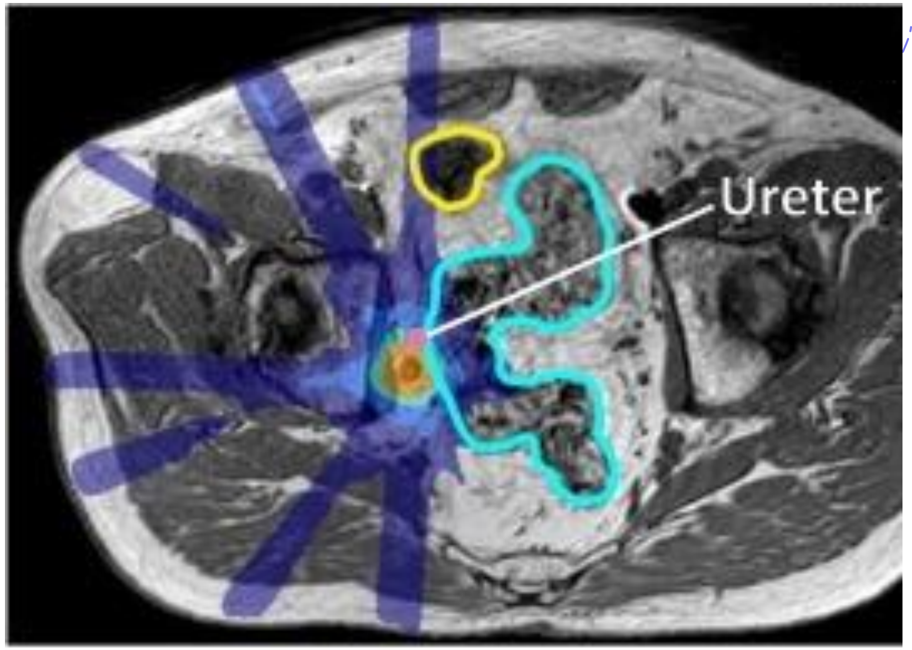
Hoegen-Saßmannshausen et al. 2025.

**Pelvic Tumors**




Zhong et al. 2023.

**Lymph nodes**



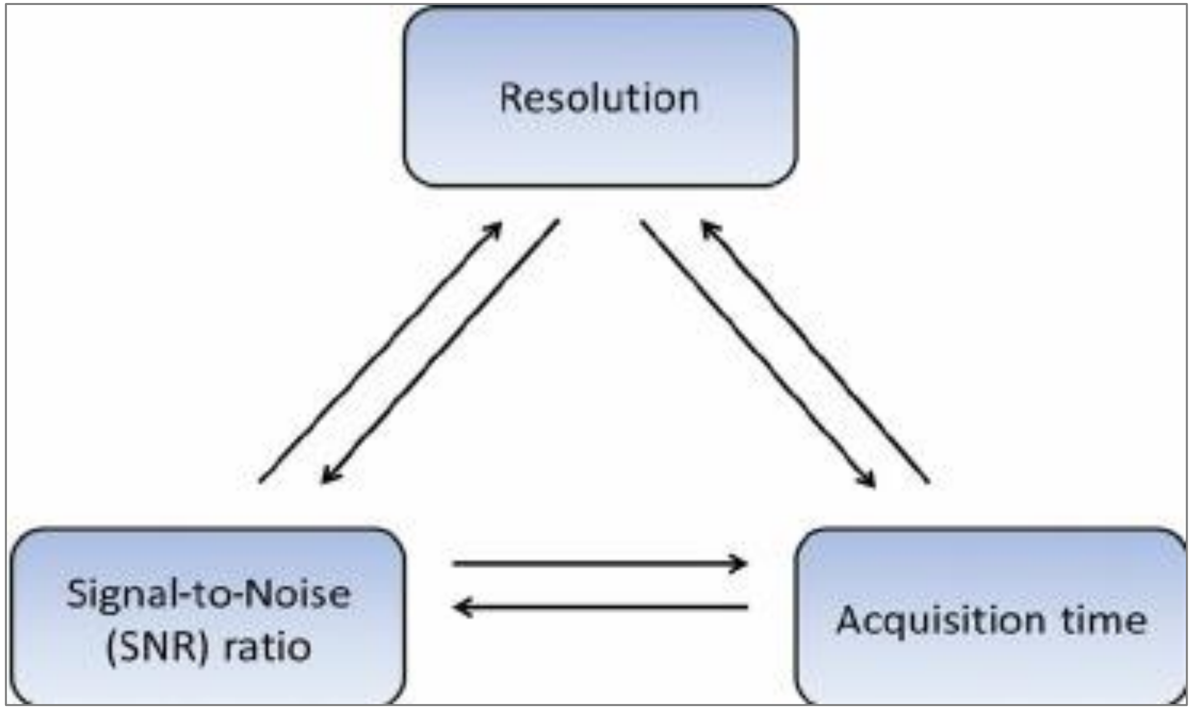

Werensteijn-Honingh et al. 2019.

# Contouring

MRI-based

- ✓ **Contouring**
- ✓ Dose calculation
- ✓ Image registration
- ✓ Artifacts

- Tradeoffs:
  - Resolution
  - Acquisition time
  - Signal-to-noise
- Careful optimization of site-specific protocols

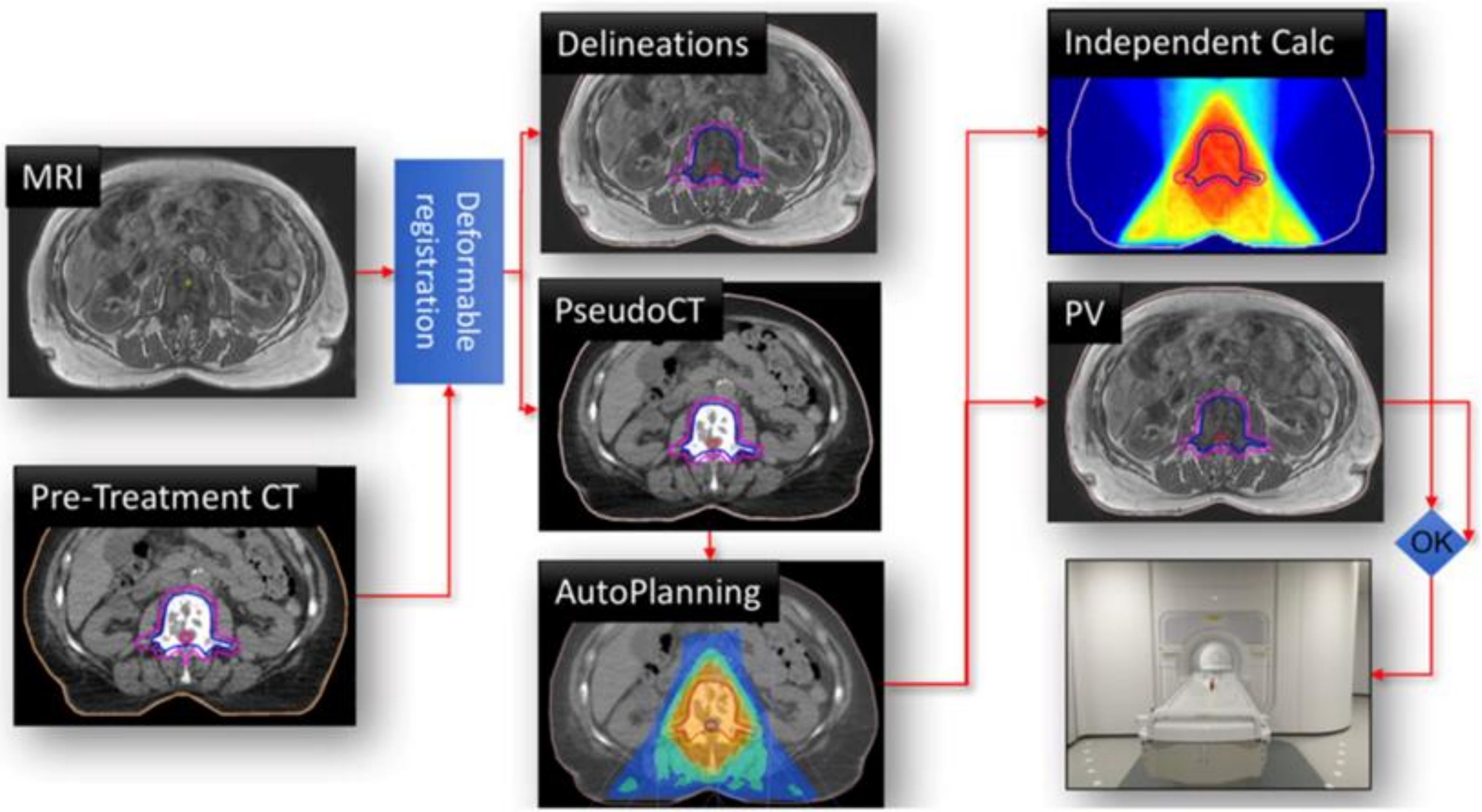



Driessen et al. 2015.

# Dose calculation


- ✓ Contouring
- ✓ **Dose calculation**
- ✓ Image registration
- ✓ Artifacts

- MRI does not provide electron density for dose calculation.
- Synthetic CTs are generated from MRIs.




Raaymakers et al. 2017.

# Dose calculation

MRI-based

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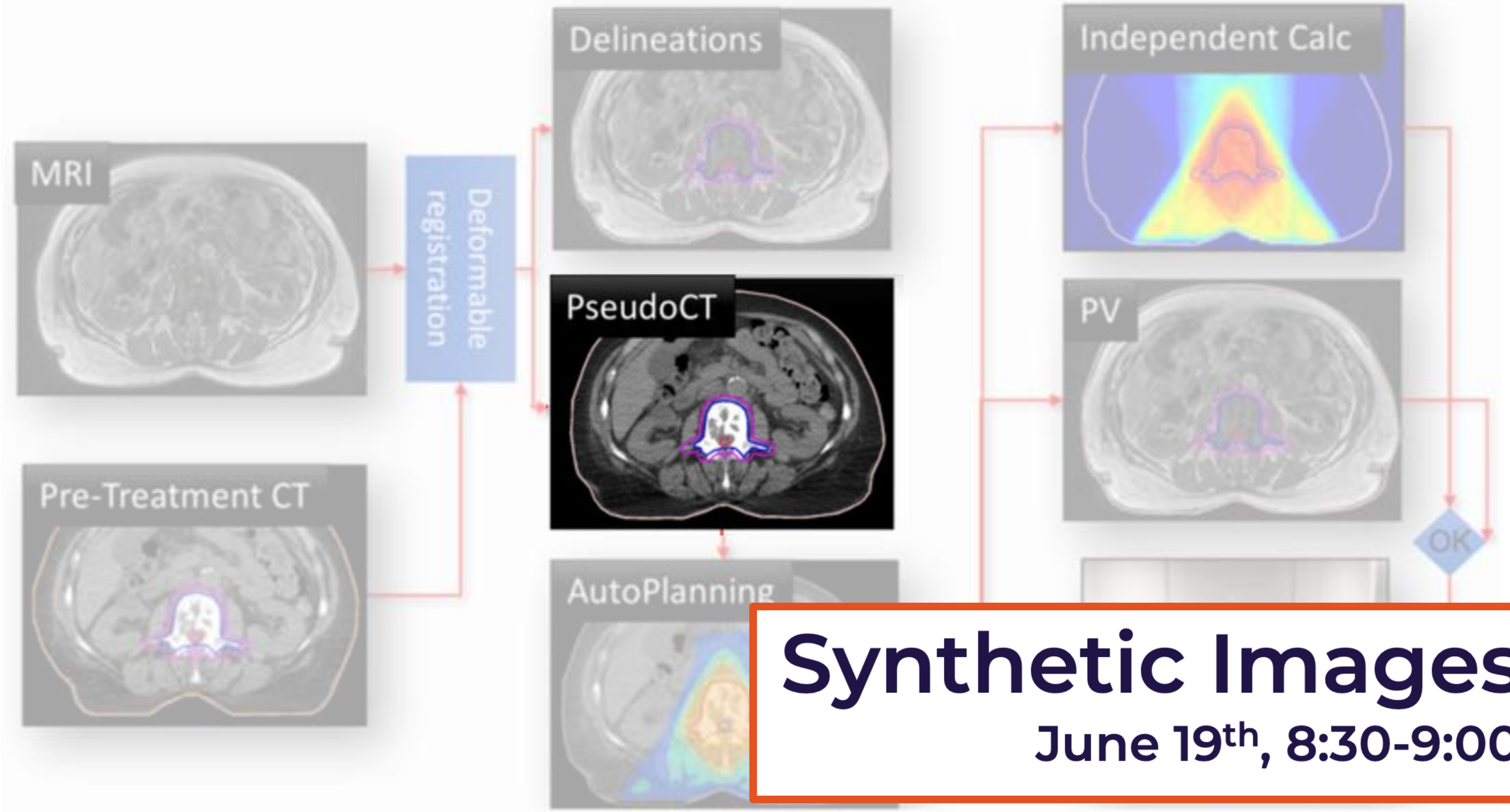



**Synthetic Images for ART**

**June 19th, 8:30-9:00am**

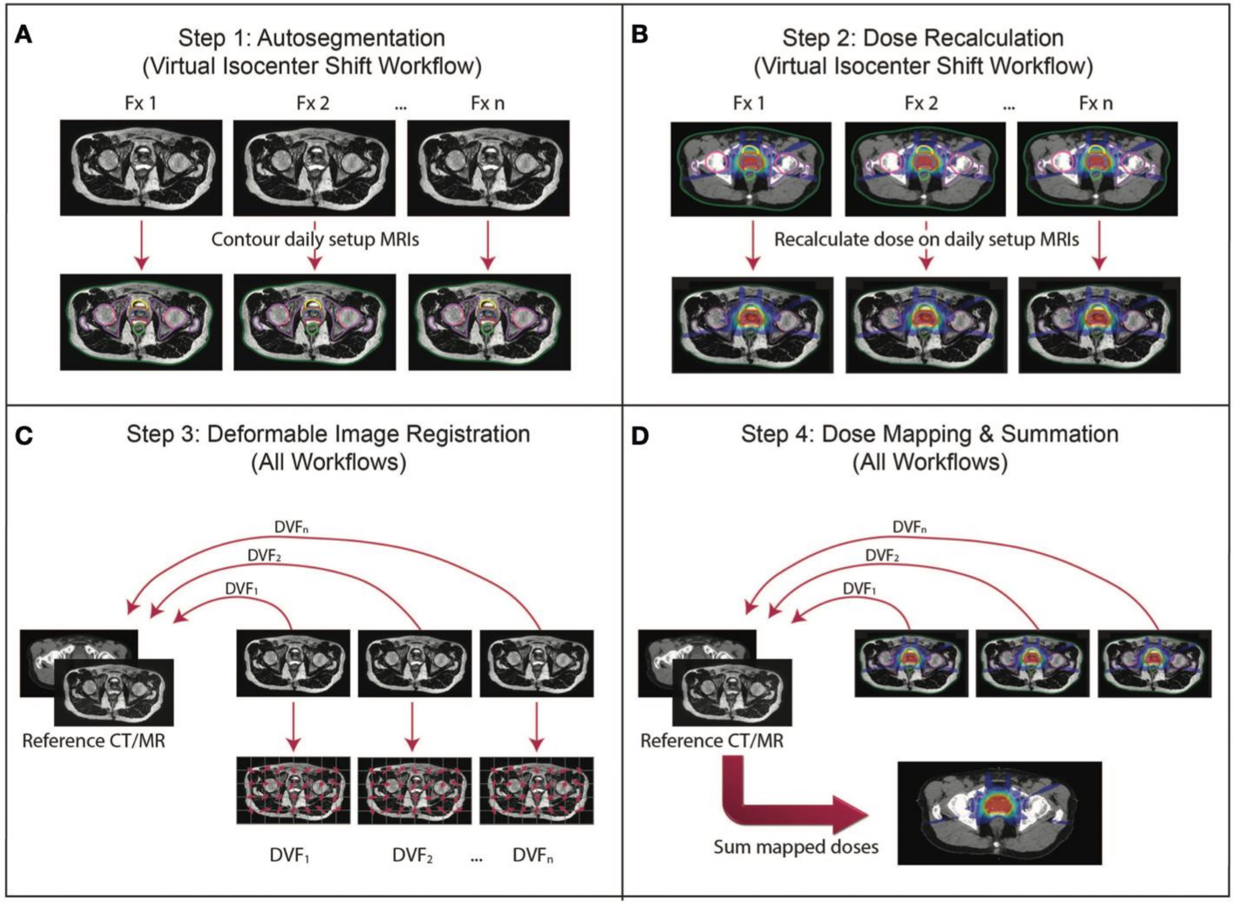
Raaymakers et al. 2017.

# Image Registration

MRI-based

- ✓ Contouring
- ✓ Dose calculation
- ✓ **Image registration**
- ✓ Artifacts

- MRIdian and Unity workflows differ
  - Adapt reference plan, adapt to position, adapt to shape, etc.
  - Use rigid or deformable registration
- Co-registration challenges in MRI
  - Signal intensity fluctuations (coils)
  - Magnetic field inhomogeneities
  - Artifacts
  - Etc.




McDonald et al. 2023.

# Artifacts – Motion

MRI-based

- ✓ Contouring
- ✓ Dose calculation
- ✓ Image registration
- ✓ **Artifacts**

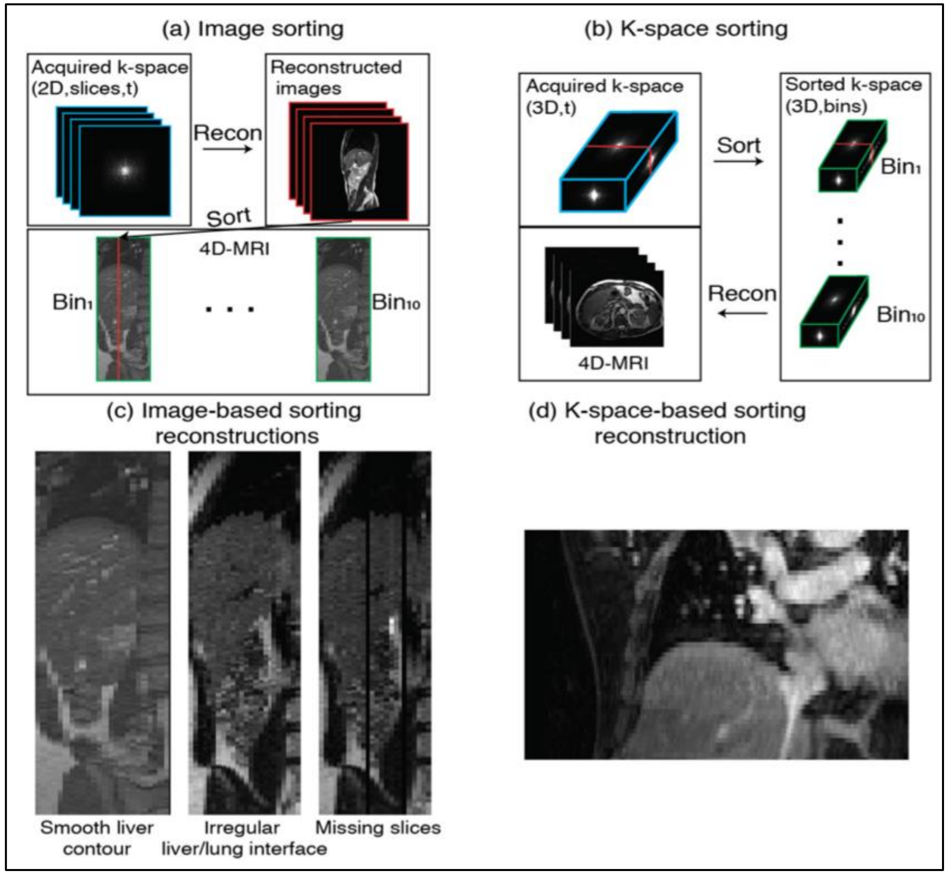
- Motion artifacts
  - Respiration
  - Gas, bowels, bladder filling, etc.

MRI-based

- ✓ Contouring
- ✓ Dose calculation
- ✓ Image registration
- ✓ **Artifacts**

# Artifacts – Motion

- Motion artifacts
  - Respiration
  - Gas, bowels, bladder filling, etc.
- How to handle
  - Understanding and mitigating implications (e.g. 2D vs. 3D MRI)
  - Compression belt
  - Bowel/bladder prep and consistency




Stemkens et al. 2018.

# Artifacts – Distortion

MRI-based

- ✓ Contouring
- ✓ Dose calculation
- ✓ Image registration
- ✓ **Artifacts**

Geometric distortions

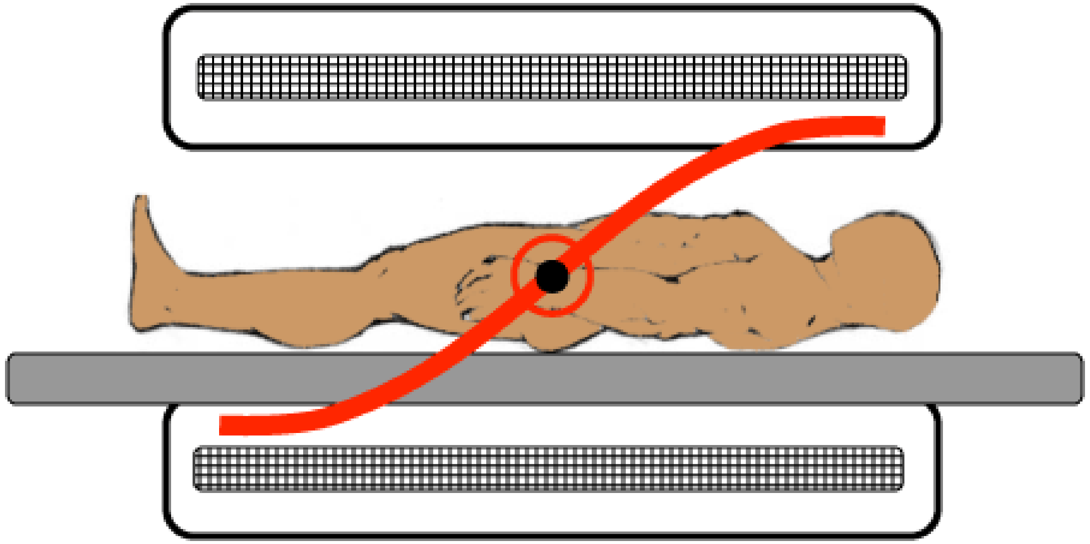
- Gradient non-linearities
- Magnetic field inhomogeneities
- Susceptibility differences (tissue, implants, etc.)

# Artifacts – Distortion


- ✓ Contouring
- ✓ Dose calculation
- ✓ Image registration
- ✓ **Artifacts**

## Geometric distortions

- **Gradient non-linearities**
  - Unavoidable hardware imperfections in gradient coils
  - Magnetic field does not change perfectly linearly with distance
  - To mitigate:
    - Apply distortion correction
    - Restrict field-of-view (FOV)
    - Position target close to isocenter



# Artifacts – Distortion


- ✓ Contouring
- ✓ Dose calculation
- ✓ Image registration
- ✓ **Artifacts**

## Geometric distortions

- **Magnetic field inhomogeneities**
  - Lack of uniformity in main magnetic field
  - Causes e.g. signal loss, distortions, fat-saturation failures
  - Worsen near susceptibility boundaries
  - To mitigate
    - Shimming
    - Use spin-echo (SE) sequences instead of gradient echo (GRE)
    - Routine QA on B0 inhomogeneities and addressing w/vendor

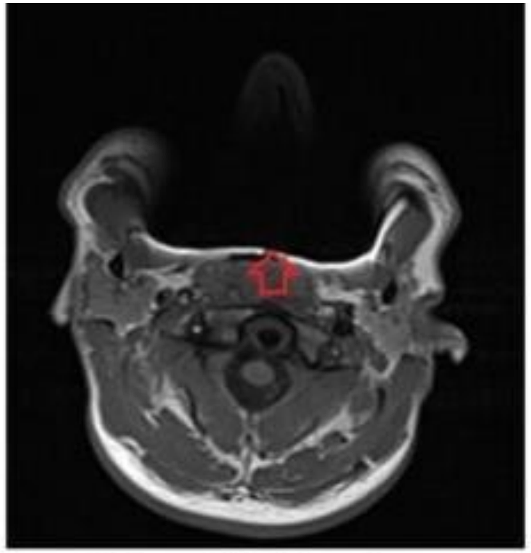
# Artifacts – Distortion

MRI-based

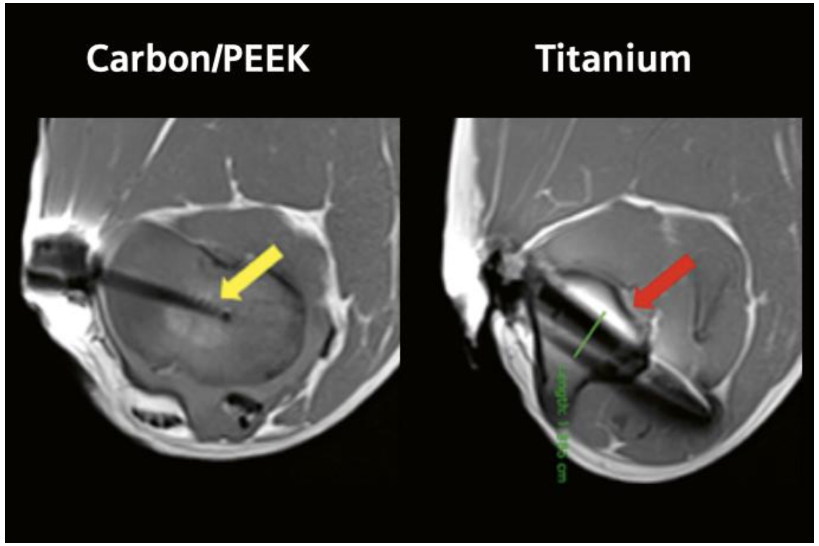
- ✓ Contouring
- ✓ Dose calculation
- ✓ Image registration
- ✓ **Artifacts**

## Geometric distortions

- **Susceptibility differences (tissue, implants, etc.)**
  - Distortion caused by substances with different magnetic properties



Metallic implant



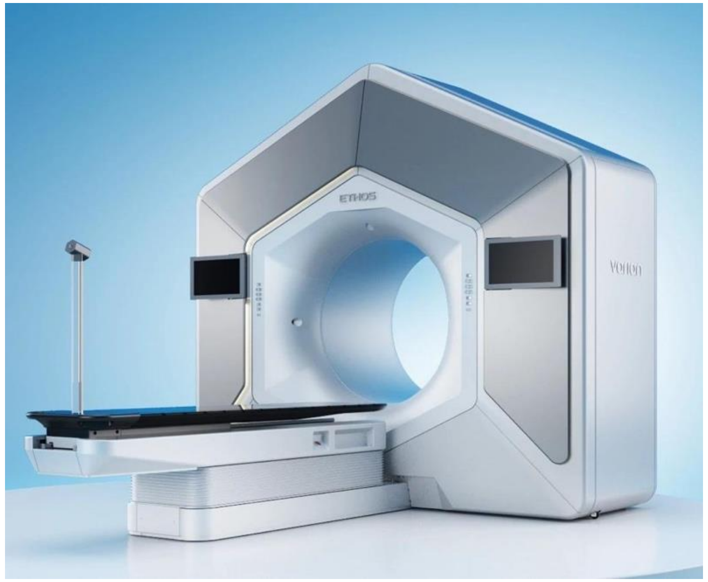

Miao et al. 2024.

- To mitigate
  - Use lower B0 field if possible
  - Use SE sequences instead of GRE
  - Increase bandwidth
  - Alter phase and frequency encoding directions
  - MR-friendly implants if possible (e.g. carbon fiber, etc.)

# Imaging Used in (online) ART

MRI-based

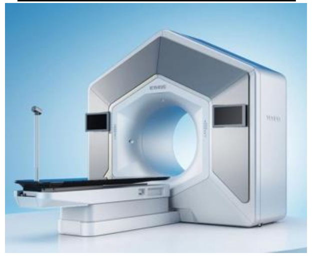
CBCT-based



BgRT-based

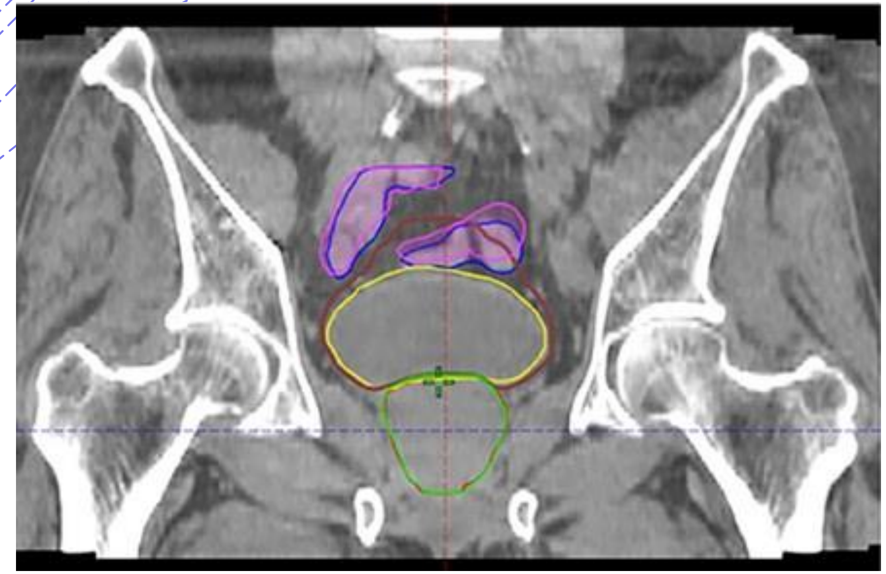
# Contouring

CBCT-based



- ✓ **Contouring**
- ✓ Dose calculation
- ✓ Image registration
- ✓ Artifacts

## Prostate Tumors

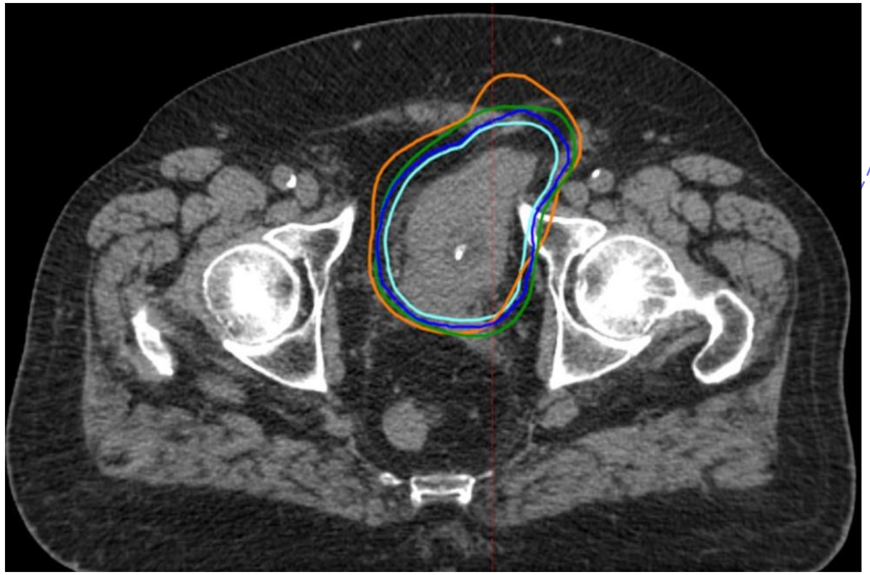


Yoganathan et al. 2025.

## Cervical Tumors

Zhang et al. 2024.

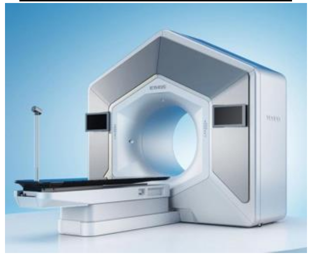
## Bladder Tumors



Åström et al. 2022.

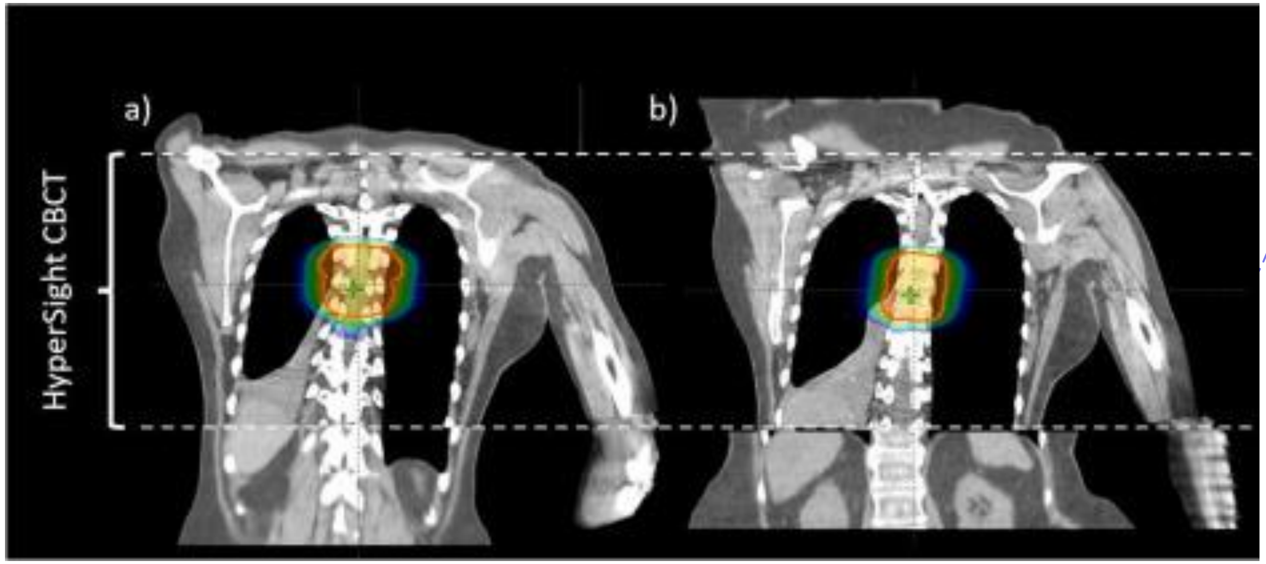
# Contouring

CBCT-based



- ✓ **Contouring**
- ✓ Dose calculation
- ✓ Image registration
- ✓ Artifacts

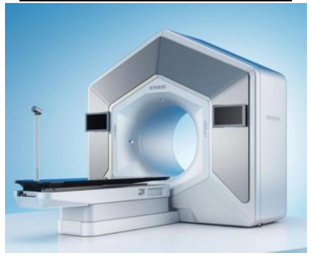
- Traditional CBCT lacks soft tissue contrast, exhibits beam hardening, scatter artifacts.
- Familiarize yourself with contouring tools
  - May be limited/different
  - Different display settings (window/level, etc.)
- Ensure important structures are included in CBCT (contours outside CBCT are propagated from reference CT).




McGrath et al. 2025.

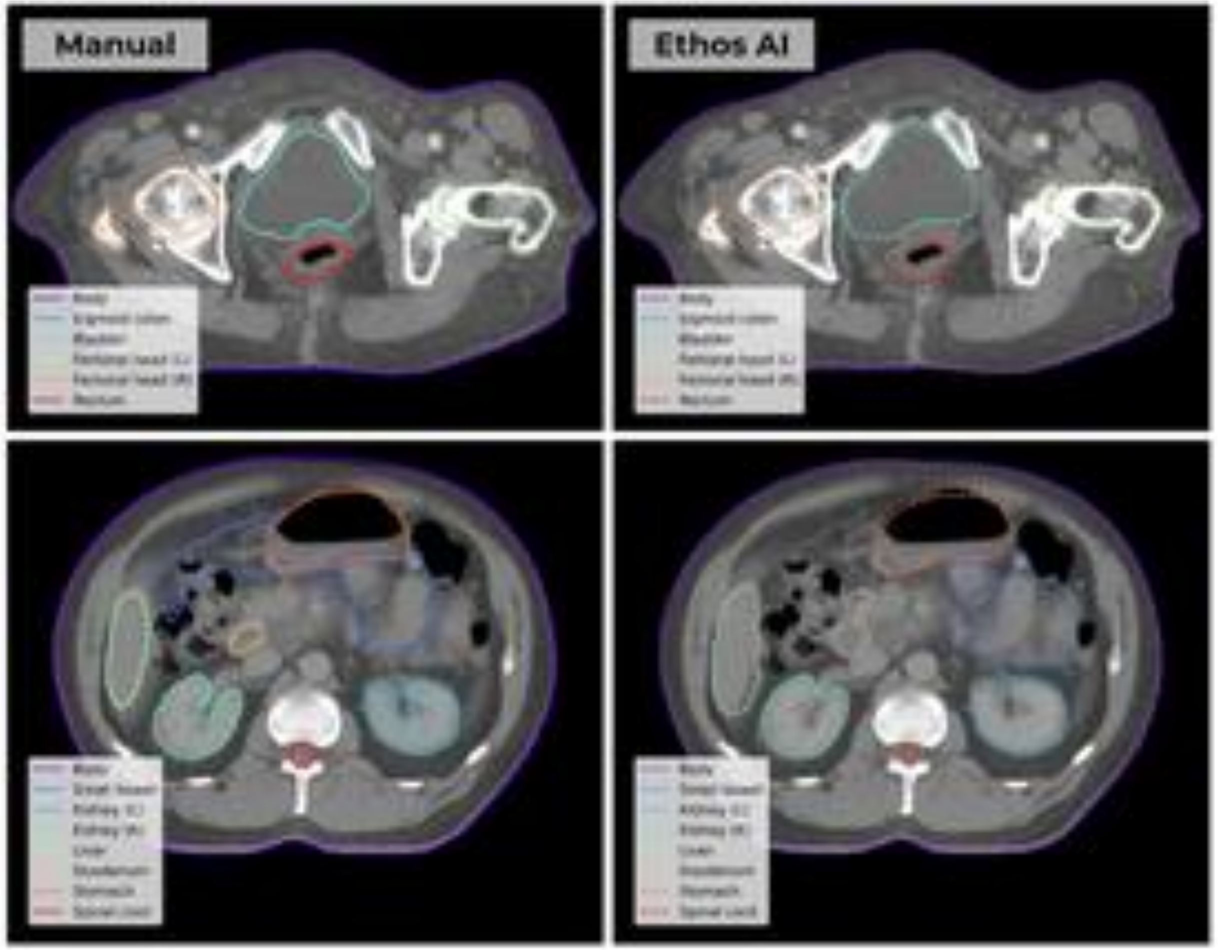
# Contouring

CBCT-based



- ✓ **Contouring**
- ✓ Dose calculation
- ✓ Image registration
- ✓ Artifacts

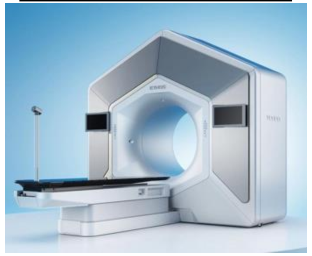
- Ethos workflow uses **auto-contouring** for OAR segmentation
  - HU-based thresholding and deep learning
  - Trained on diverse dataset, post-processing used to smooth, fill cavities, etc.
- Auto- and manual contours should be reviewed slice-by-slice.
  - Consider clinical implications of contour accuracy (balance safety, efficacy, and efficiency).




Finnegan et al. 2025.

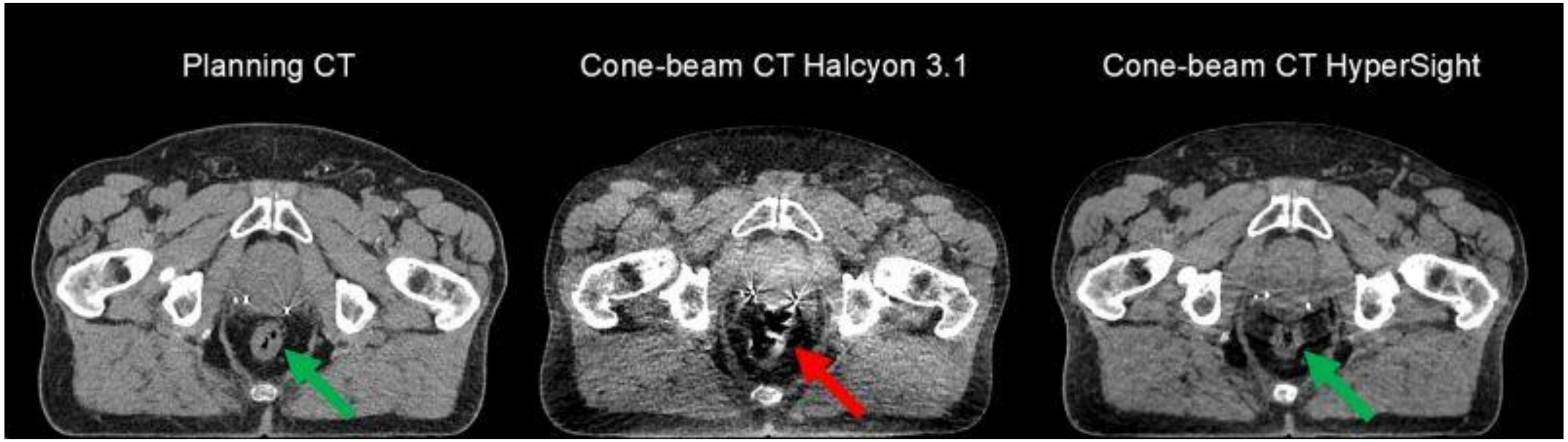
# Contouring

CBCT-based



- ✓ **Contouring**
- ✓ Dose calculation
- ✓ Image registration
- ✓ Artifacts

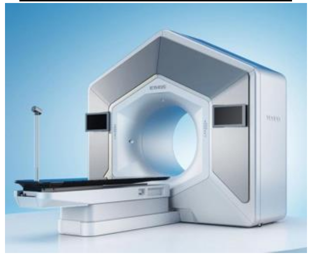
- Upgraded Ethos features **HyperSight**
  - Extended field-of-view (53.8cm → 70cm)
  - Detector upgrades and larger panels
  - Improvements to reconstruction, resolution, metal artifact reduction, etc.
  - Rapid acquisition and reconstruction
  - Reduces soft-tissue blur that typically hinders AI segmentation algorithms.




Erasmus MC Cancer Institute. 2023 (online).

# Dose calculation

CBCT-based



- ✓ Contouring
- ✓ **Dose calculation**
- ✓ Image registration
- ✓ Artifacts

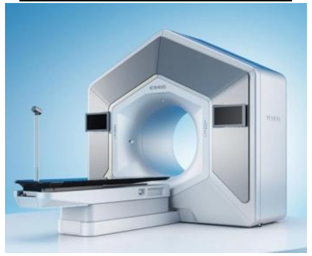
- Standard CBCTs → artifact noise, inconsistent HU stability
  - Cannot be used for direct dose calculation.
- Ethos resolves through registration-based density mapping:
  - Planning CT deformed to CBCT for online dose calculation.
- Hypersight offers direct dose on CBCT

**Synthetic Images for ART**

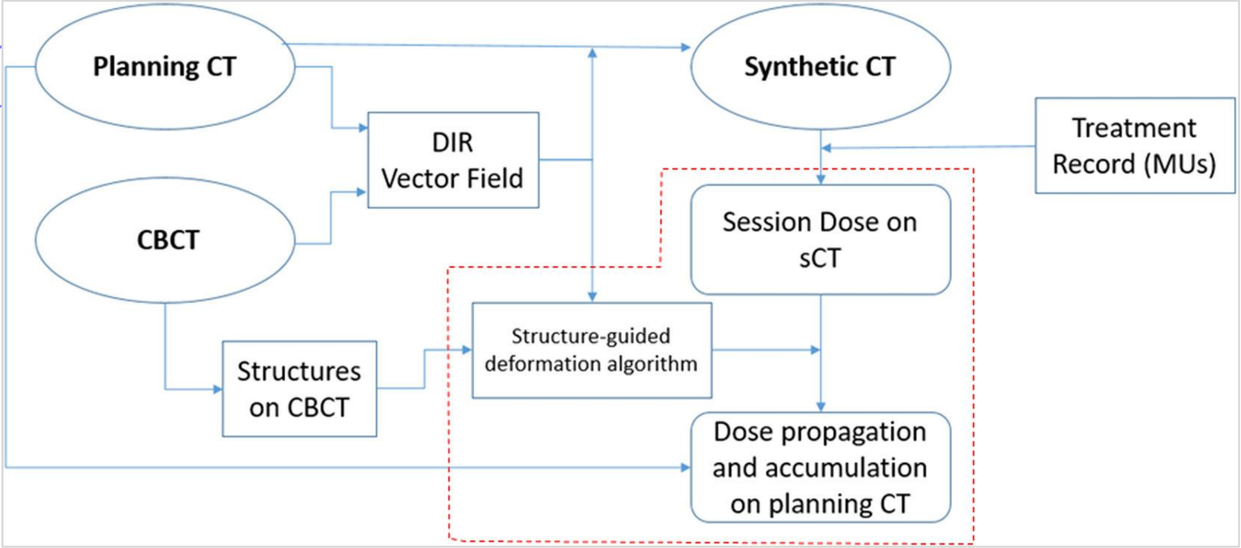
**June 19th, 8:30-9:00am**

# Image registration (Ethos)

CBCT-based



- ✓ Contouring
- ✓ Dose calculation
- ✓ **Image registration**
- ✓ Artifacts



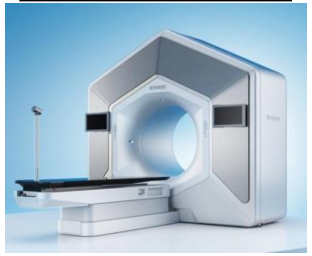

Maraghechi et al. 2023.

Image registration is multi-step process:

- **Rigid** registration (translation + rotation)
  - Couch shifts determined from translations only
- **Deformable** registration
  1. Image-guided DIR
     Elastic deformation for CT-sim to CBCT
  2. Structure-guided DIR
     Structure propagation and dose accumulation

# Image registration

CBCT-based



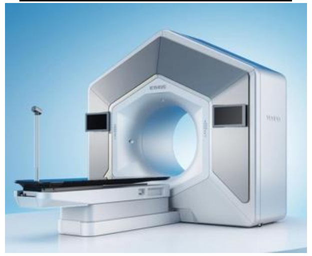
- ✓ Contouring
- ✓ Dose calculation
- ✓ **Image registration**
- ✓ Artifacts

Image features impacting registration

- Structural boundaries
  - Edge alignment optimization
- Truncation of body contour
  - DIR algorithm attempts to compress or stretch the reference planning CT
- Structure-guided DIR impacted by influencer and target structures
- Artifacts

# Artifacts – metal, air

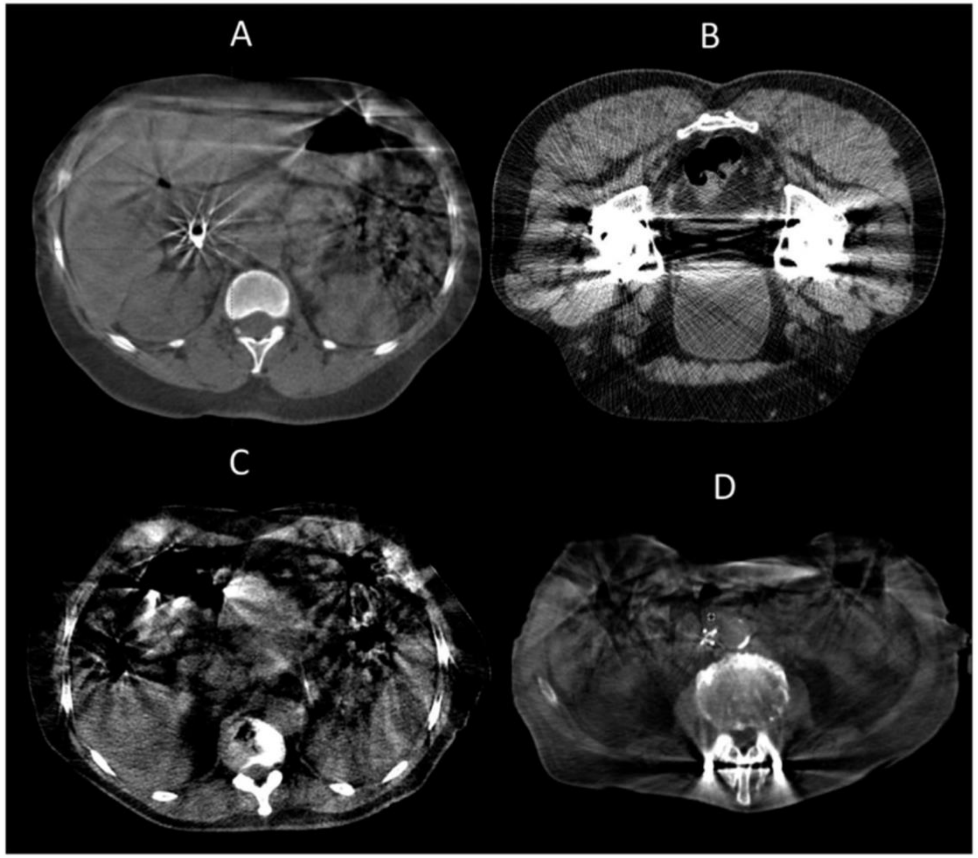
CBCT-based



- ✓ Contouring
- ✓ Dose calculation
- ✓ Image registration
- ✓ **Artifacts**

- Streaking artifacts on CBCT caused by
  - Hip prostheses
  - Dental implants
  - Spinal hardware
  - Etc.

- To mitigate:
  - Utilize metal artifact reduction (MAR).
  - Can apply proactive manual overrides on planning CT (propagated to daily sCT automatically).
    - Use cautiously as all overrides on planning CT are propagated forward!

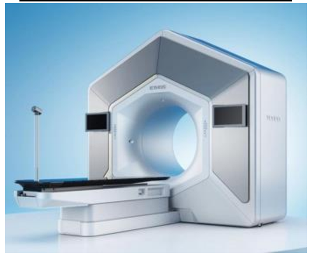



Ethos CBCT artifacts: (A) biliary stent, (B) hip prosthesis, (C) gas pockets, and (D) spine implants can affect online target and OAR delineation.

Wang et al. 2023.

# Artifacts – motion

CBCT-based

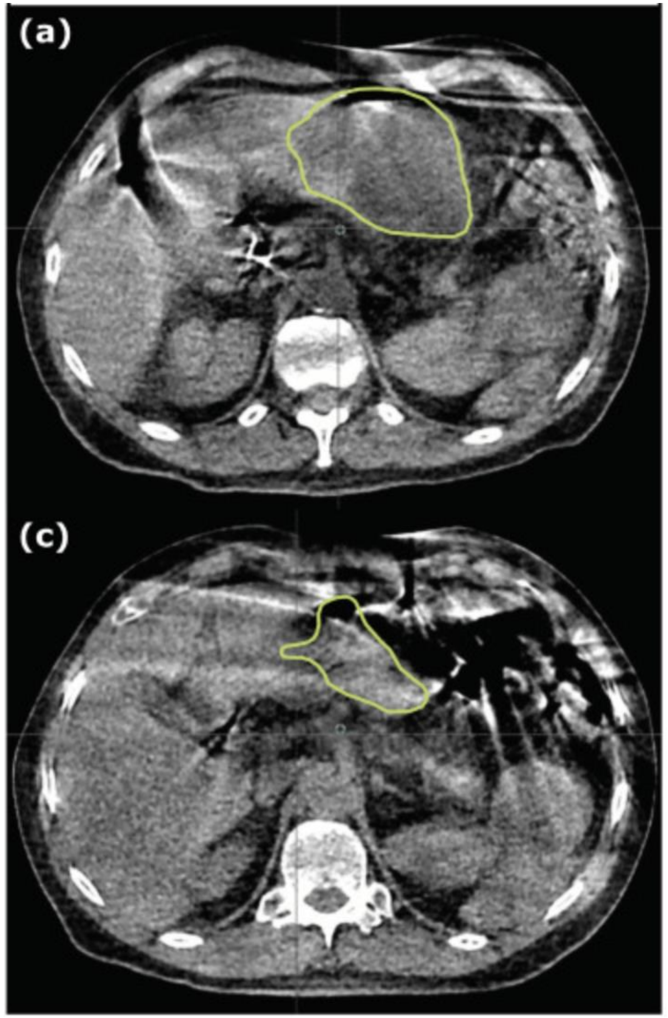


- Contouring
- Dose calculation
- Image registration
- **Artifacts**

- Motion artifacts can be detrimental to contouring.
  - Abdominal targets: **breath-hold** or **fast-acquisition** to avoid severe distortions.
  - Daily variations in rectal gas, bowel distension, or bladder filling may introduce local contrast anomalies.
  - Consider **pre-treatment bowel preparation** protocols.
  - To mitigate gas bubble artifacts, consider repeat CBCT.


THE UNIVERSITY OF ALABAMA AT BIRMINGHAM

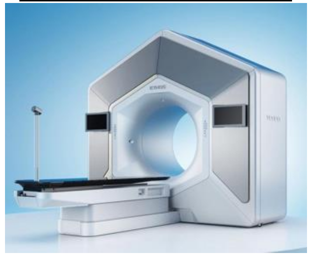
Gill et al.




Olberg et al. 2025.

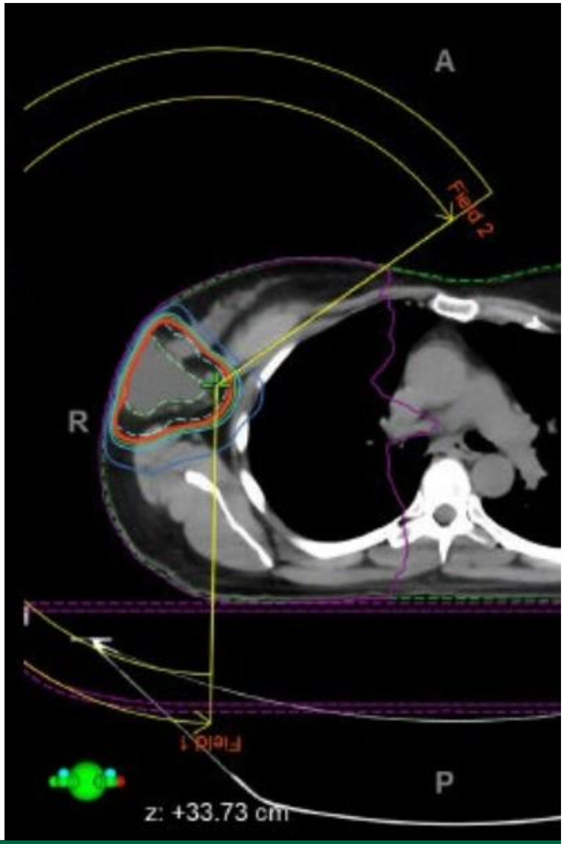
# Artifacts – limited FOV

CBCT-based



- ✓ Contouring
- ✓ Dose calculation
- ✓ Image registration
- ✓ **Artifacts**

- Use **Extended FOV** reconstruction to minimize truncation.
- Limited FOV is accounted for by stitching CT-sim sup-inf but not axially.
- Be cautious with beam angles if CBCT is cut-off laterally.
- Avoid beams entering through missing tissue.




Pogue et al.

# Imaging Used in (online) ART

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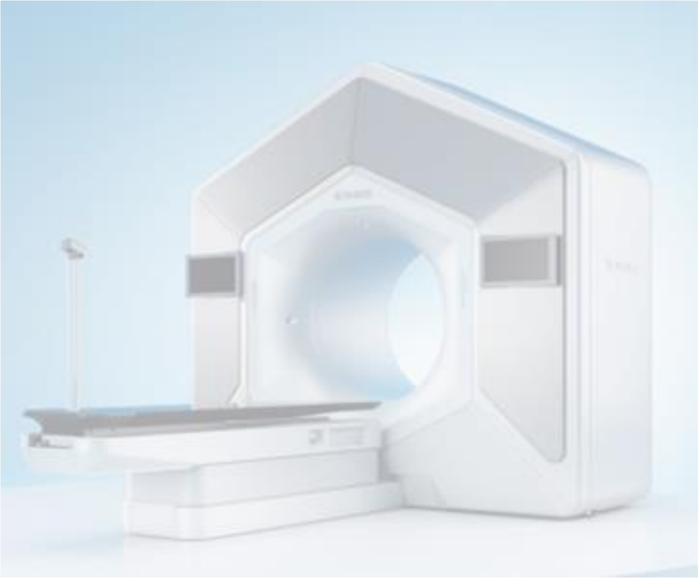
CBCT-based



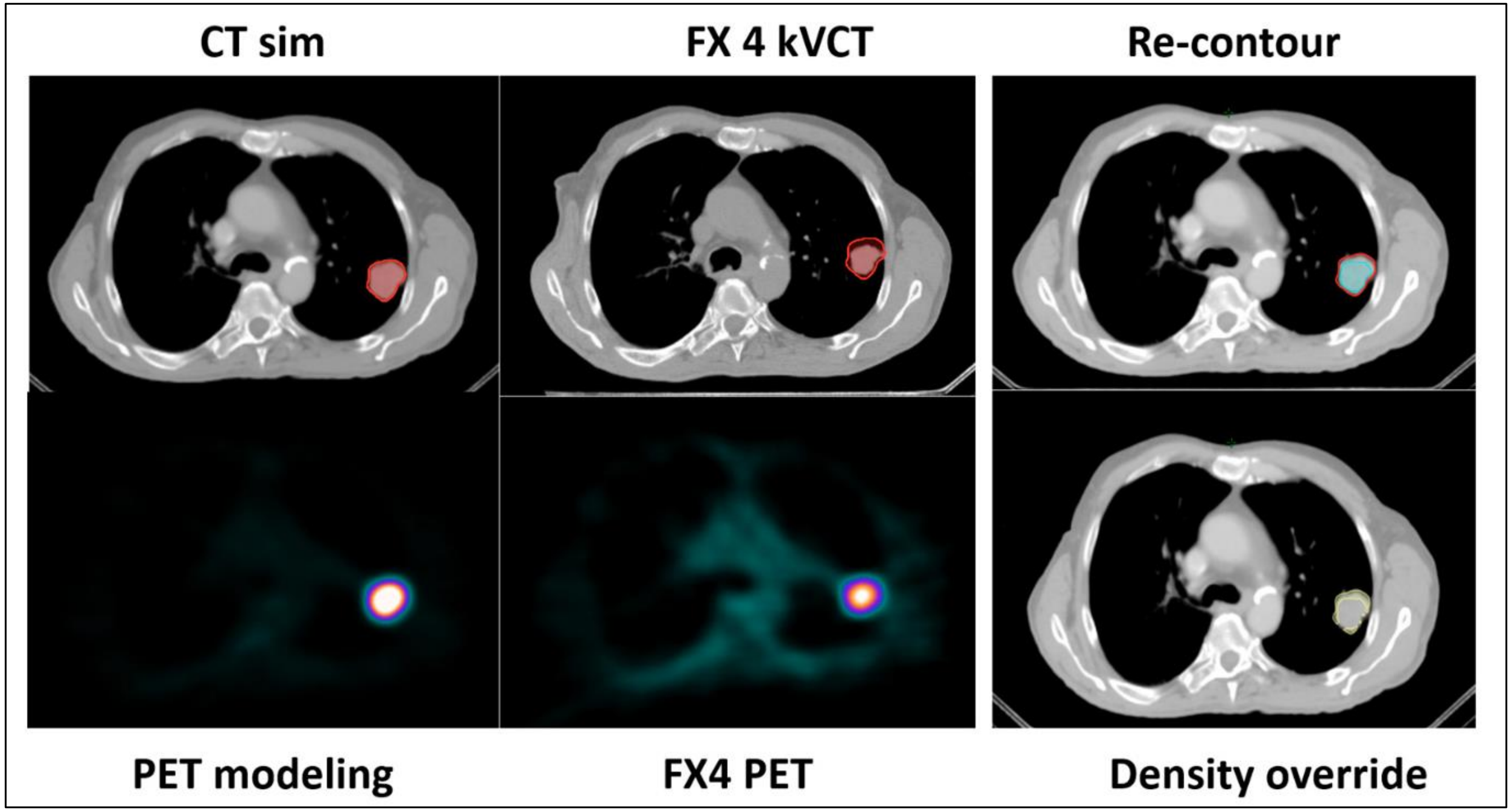
BgRT-based

# Contouring

BgRT-based

- ✓ **Contouring**
- ✓ Dose calculation
- ✓ Image registration
- ✓ Artifacts

RefleXion: integration of fan-beam kilovoltage CT (**kVCT**) and **PET** biology-guided RT (BgRT).



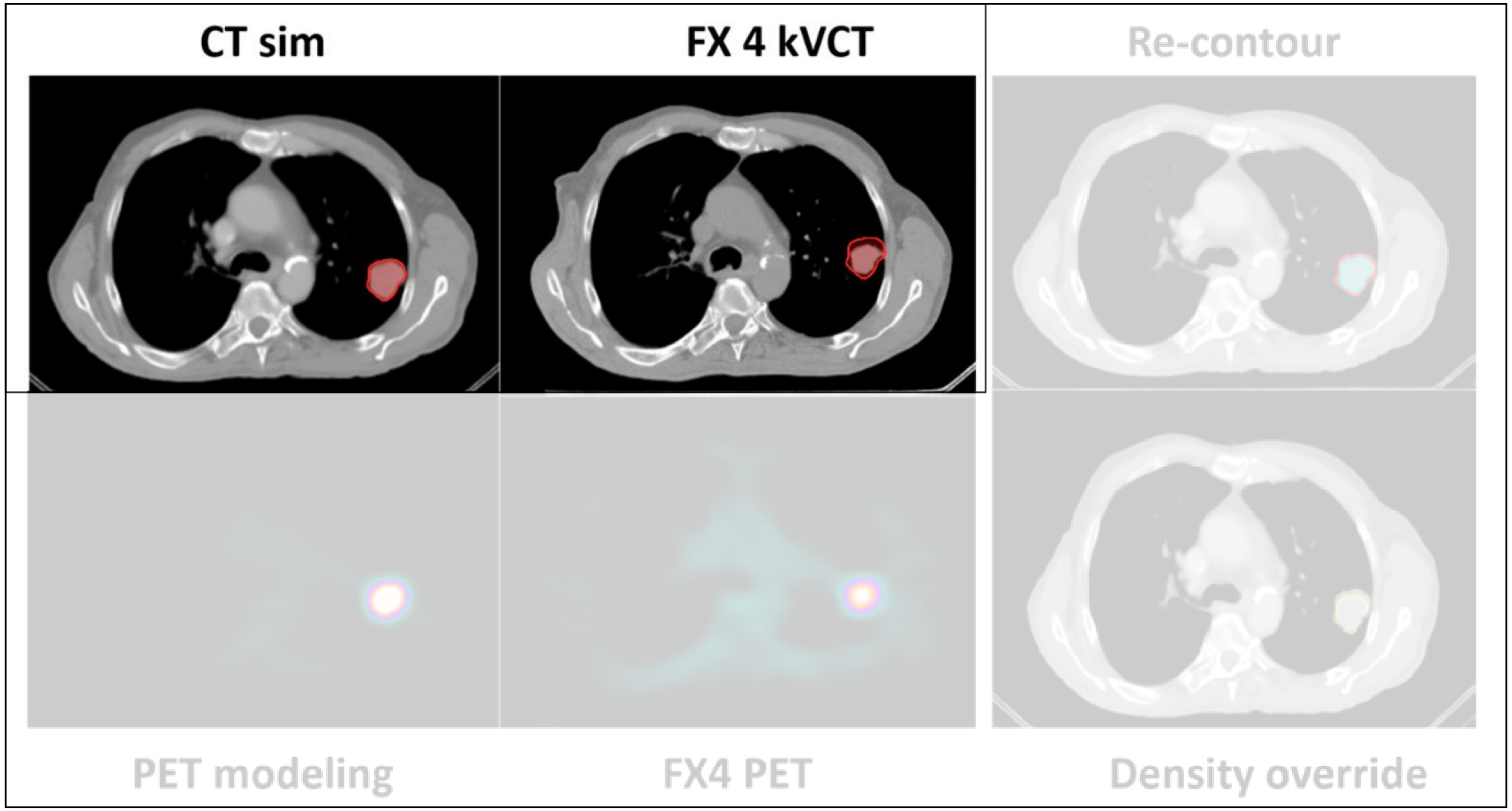

Cai et al. 2025.

# Contouring

BgRT-based

- ✓ **Contouring**
- ✓ Dose calculation
- ✓ Image registration
- ✓ Artifacts

Fan-beam kVCT has significantly **higher contrast-to-noise, higher spatial resolution**, and **fewer motion artifacts** than a conventional CBCT.




Cai et al. 2025.

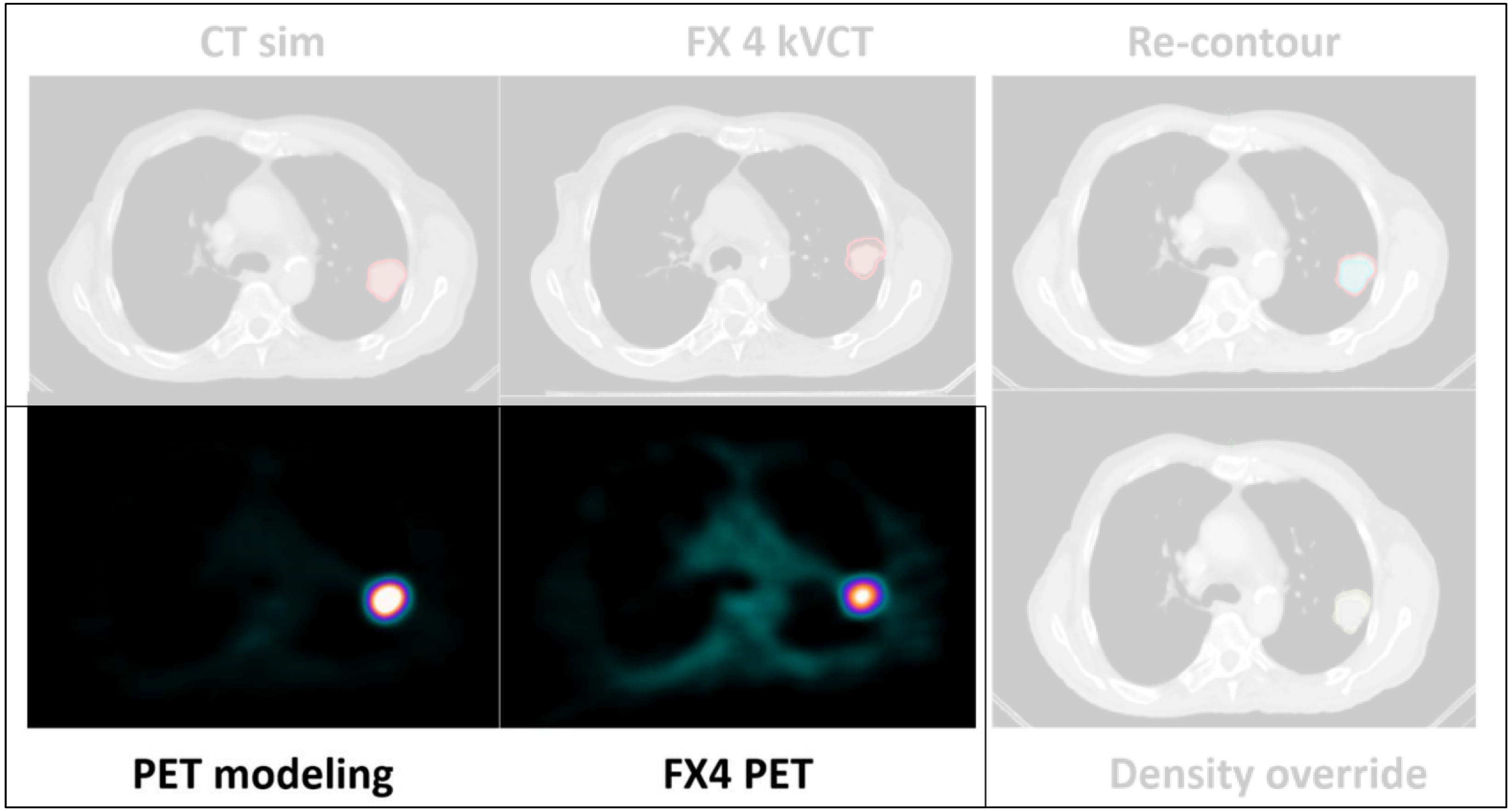
# Contouring

BgRT-based

- ✓ **Contouring**
- ✓ Dose calculation
- ✓ Image registration
- ✓ Artifacts

On-board PET detectors track continuous stream of positron emissions from the **radiotracer (FDG).**

- SCINTIX® therapy FDA-cleared for BgRT-guided treatment of <u>lung</u> and <u>bone</u> tumors (2023).




Cai et al. 2025.

# Contouring

BgRT-based

- ✓ **Contouring**
- ✓ Dose calculation
- ✓ Image registration
- ✓ Artifacts

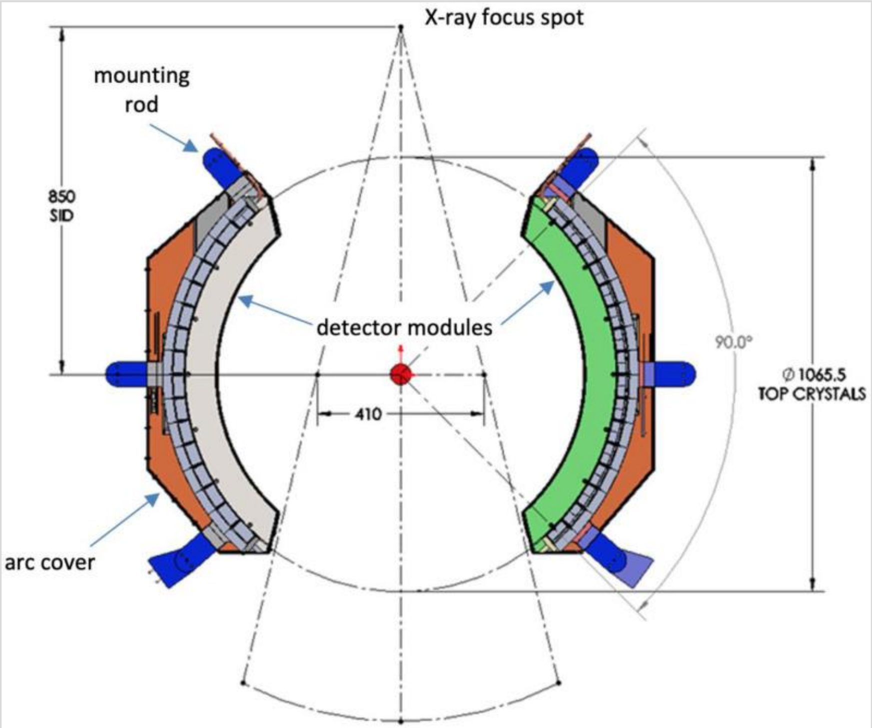
- Physician defines Biology-guided Target Volume (BgTV) (offline)
  - High metabolic signals from highly concentrated regions (e.g. urinary catheter, bladder wall, kidneys) may spillover.
  - Tracking zone set around ITV. Other high uptake structures excluded.
- During delivery, machine:
  - Calculates tumor's location based on live metabolic activity.
  - Dynamically steers beam dynamically.

# Contouring (X1 System)

BgRT-based

- ✓ **Contouring**
- ✓ Dose calculation
- ✓ Image registration
- ✓ Artifacts

- **Tumor Size Eligibility**:
  - >2cm (partial volume effect)
  - <5.1cm (axial FOV of PET detector)
- **Imaging mode**
  - Spatial resolution, image contrast comparable to diagnostic PET.
  - Sensitivity, count rate lower due to small PET detector area.
- **Treatment mode** (real-time tracking)
  - Motion tracking limited by ~1.2cm in each direction.
  - Balance between signal and latency → noisy images.
  - Tradeoff in post-processing time for (back-filtered reconstruction)




Hu et al. 2023.

*X2 system increases FOV to 20cm, increases PET sensitivity.*

*(510(k) cleared early-2026)*

# Dose calculation

BgRT-based

- ✓ Contouring
- ✓ **Dose calculation**
- ✓ Image registration
- ✓ Artifacts

- RefleXion system equipped with 16-slice diagnostic-grade fan-beam kVCT.
- Planning CT still used as reference for:
  - Dose calculation and accumulation
  - CT-based contouring
  - PET attenuation correction
- Assume patient is static, daily kVCT is co-registered to CT-sim reference image and all session images are co-registered to CT-sim and dose is accumulated there.

# Image registration

BgRT-based

- ✓ Contouring
- ✓ Dose calculation
- ✓ **Image registration**
- ✓ Artifacts

- Planning CT to daily kVCT
  - Offline registration
  - Recent upgrade allows for automatic DIR
- Daily PET to daily kVCT
  - Simultaneous scanning to establish structural-to-metabolic coordinates.

# Artifacts - kVCT

BgRT-based

- ✓ Contouring
- ✓ Dose calculation
- ✓ Image registration
- ✓ **Artifacts**

- Fast gantry rotation (60 rpm) mitigates motion blurring, captures sharp anatomical features relative to CBCT.
    - **Rapid / irregular respiratory movement** or **sudden digestive tract gas** movement can cause slice-to-slice displacement artifacts.
    - For ultra-precise SBRT targets, consider breath-hold.

# Artifacts - PET

BgRT-based

- ✓ Contouring
- ✓ Dose calculation
- ✓ Image registration
- ✓ **Artifacts**

- **Functional image quality** dictates dynamic beam targeting
  - Biological tracking highly sensitive to background radiotracer accumulation.
  - **Poor metabolic uptake** or **high local noise** may cause real-time image quality to drop below these safety thresholds

    → Resort back to standard anatomical margins

- Ensure **consistency in injection timing and activity** (nuclear medicine may accept higher variability).
  - Consider using auto-injection rather than manual injection.
  - Regimented timing of appointments (injection, waiting, scan acquisition, etc.).

# Image Quality Summary

| | MR-based | CBCT-based | BgRT |
|---|---|---|---|
| Tissue contrast | Superior soft tissue contrast | Lowest contrast | PET-avid contrast |
| Electron density (ED) | Poor ED accuracy (synthetic CT) | Medium ED accuracy (correct using sCT, DIR) | High ED accuracy |
| Image registration (dose accumulation) | Intensity-based DIR | Structure-guided DIR | Planning CT-to-daily CT (dose on planning CT) |
| Artifacts | Susceptibility, motion, distortion | Beam hardening, motion, scatter | Metabolic and tracer kinetics distortions, motion |
| Geometric distortions | Present* | N/A | N/A |
| Limits on body size | More restrictions | Low/medium restrictions | Low restrictions on kVCT<br>High restrictions on PET |
| Imaging exposure | None | Lower (~0.2-2cGy/scan) | Higher (~1-3 cGy/CT scan, ~5-7 mSv/PET scan) |
| Continuous imaging | Possible | Limited by radiation | Limited by radiation |
| Imaging time | Long | Short | Long (PET) |
| Functional imaging | Possible | N/A | Yes |

*Geometric distortions in MRI can be caused by gradient non-linearities, magnetic field inhomogeneities, and patient-specific distortions.

# Image Quality Considerations

## *Breakout Table*

June $18^{th}$ | 1:15-2:15pm

Jess Scholey PhD DABR
University of California San Francisco

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